



Fungal diversity associated with coffee leaf rust (*Hemileia vastatrix*) pustules based on ITS1 amplicon sequencing

Dirk Berkelmann^{1,4} · Juan Miguel Zuñiga-Umaña¹ · Priscila Chaverri² · William Solano³ · Andrés Gatica-Arias¹

Received: 8 August 2025 / Accepted: 11 March 2026
© The Author(s), under exclusive licence to Springer Nature B.V. 2026

Abstract

Coffee leaf rust (CLR), caused by *Hemileia vastatrix*, is one of the biggest economic challenges for coffee cultivation and leads to high economic losses each year. Co-occurring fungal microbial communities and their diversity in the presence of CLR are widely understudied but may harbor potential agents or indicators to reduce CLR infections. In this study, the fungal communities associated with CLR pustules in *Coffea arabica* L. plants across different regions of Costa Rica were analyzed. To this end, individual pustules were excised from infected leaf tissue and used as source material for DNA extraction and subsequent amplification and sequencing of the fungal taxonomic marker region ITS1. Effects of altitude and location on fungal community structure were also observed. High taxonomic variance within regions and a large proportion of unclassified taxa were detected as well as similar community structures across regions, possibly reflecting small effects of the analyzed regions on the identified taxa. However, altitude was a significant factor on the detected community structure, indicating either less favorable growth conditions for the pathogen in higher regions or favorable conditions for co-occurring taxa. This emphasizes that taxonomic identification of co-occurring fungi and their ecological relevance (e.g., potential mycoparasites) during CLR infection requires further research. This study provides a foundational framework for global coffee research by emphasizing the untapped potential of fungal community analyses to develop innovative, microbiome-informed strategies for managing coffee leaf rust and improving crop resilience.

Keywords Microbiome · Co-occurrent communities · Coffee leaf rust · Metabarcoding · Microbial ecology

Introduction

Coffee agroecosystems, which hold substantial economic and ecological importance worldwide, provide valuable systems for exploring the structure and function of plant-associated microbiota (Jurburg et al. 2020). In this context, fungal communities in Costa Rican coffee plantations exhibit remarkable complexity and contribute fundamentally to

plant health, productivity, and ecosystem stability (Schwan et al. 2012; Bilen et al. 2023). Among the diseases that affect coffee production, coffee leaf rust (CLR), caused by the biotrophic fungus *Hemileia vastatrix* Berk. & Broome, represents one of the principal causes of yield and productivity losses worldwide (Avelino et al. 2015; Hajian-Forooshani et al. 2016; Talhinhos et al. 2017; Gichuru et al. 2021; Resende et al. 2021; Silva et al. 2022; Koutouleas et al. 2023). Due to the tropical climate and thus intensive periods of precipitation, rust fungi pose a considerable risk to agriculture in general in Costa Rica (Talhinhos et al. 2017; García-Nevarez et al. 2025). Strong regional differences in elevation among other factors create a wide variety of microclimates that also play a role in the distribution of pathogens and niches for proliferation (Talhinhos et al. 2017; Castillo et al. 2020; Gichuru et al. 2021; Resende et al. 2021).

The development of CLR is favored by temperatures between 10 °C and 35 °C and high relative humidity, creating ideal conditions for the growth and spread of *Hemileia vastatrix*. Consequently, tropical regions provide especially

✉ Andrés Gatica-Arias
andres.gatica@ucr.ac.cr

¹ University of Costa Rica, School of Biology, Plant Biotechnology Laboratory, San José, Costa Rica

² Department of Natural Sciences, Bowie State University, Bowie, MD, United States of America

³ CATIE (Centro Agronómico Tropical de Investigación y Enseñanza), Turrialba, Cartago, Costa Rica

⁴ Tropical Diseases Research Program, School of Veterinary Medicine, Universidad de Heredia, Heredia, Costa Rica

suitable environments for the establishment and epidemic progression of leaf rust fungi (Castillo et al. 2020; Gichuru et al. 2021; De Oliveira Aparecido et al. 2024). Infection causes orange spots on the underside of the leaf after an incubation time of about 48 h (Talhinhas et al. 2017; Daba et al. 2024). As the disease progresses, lesions expand and their centers undergo necrosis, ultimately resulting in severe leaf damage and plant defoliation (Gichuru et al. 2021). This in turn, affects plant growth and thus cultivation for fruits due to a decrease in photosynthesis rates. As a critical constraint for coffee production, CLR can result in severe economic losses ranging from 30% to 90%, with South America being among the most affected regions due to its conducive environmental conditions for disease development (Resende et al. 2021; Koutouleas et al. 2023).

Several promising approaches have been developed to mitigate the economic losses caused by rust fungi, including the use of genetically resistant cultivars, fungicide application, biological control, disease forecasting, and emerging RNA-based technologies (Degnan et al. 2023; Poma-Angamarca et al. 2024). Beyond these chemical and breeding strategies, growing attention has been directed toward the role of coffee-associated microbiota and microbial communities, which represent a particularly promising path for sustainable disease management (Hajian-Forooshani et al. 2016; Duong et al. 2020). Next to cultivation and application of resistance-enhancing or counteracting organisms, it is vital to understand microbial community development during the actual infection as well. As fungi produce a wide variety of metabolites and agents that aid in the defense against pathogenic fungi, studying fungal diversity and their ecological roles is a keystone for fully understanding the interspecific processes involved in rust infections (Strobel 2018; Tedersoo et al. 2020). Such interactions between rust fungi and other microorganisms underscore the complexity of fungal infections in coffee cultivation. Additionally, abiotic factors such as temperature, precipitation, shade cover, and altitude can be factors for CLR infections (Groenen 2018; Castillo et al. 2020; Zewdie et al. 2020; Anhar et al. 2021).

Plant microbiota can be studied using various approaches, including culture-dependent and culture-independent methods. The most common approaches are either by cultivation of single isolates from plant material or by metagenomic sequencing approaches by using taxonomic marker genes like ITS1/2 or metagenomic shotgun sequencing with DNA extracts from plant tissue or surrounding soil (Oliveira et al. 2014; Duong et al. 2020; Fulthorpe et al. 2020; Wemheuer et al. 2020; Sousa et al. 2021; Bez et al. 2023; Poma-Angamarca et al. 2024). To analyze the full communities, high-throughput sequencing (HTS) approaches are more fitting but are highly dependent on the completeness of

databases to classify obtained sequences (Klindworth et al. 2013; Breitwieser et al. 2019). Especially for fungi, a high degree of unclassified taxa is still a challenge for studies aiming for metagenomic communities in plants (Tedersoo et al. 2020; Gołębiewski and Tretyn 2020; Thomas et al. 2020; Phukhamsakda et al. 2022). Previous studies investigated the shape of microbial communities in rhizosphere, soil, or even the surface of coffee plants, but rarely the endophytic communities in the context of an active pathogen infection (Schwan et al. 2012; Resende et al. 2021; Bilen et al. 2023). Most of the previous studies utilized a culture-dependent approach, in which fungal or bacterial isolates were grown in culture and analyzed for effects on CLR infection (Lu et al. 2022; Luiz et al. 2024). While this allows studying growth behavior and interspecific effects of the obtained isolates, overall biodiversity and community structures in the context of CLR cannot be addressed since the vast majority of microorganisms cannot be cultured (Laval et al. 2021).

This study aimed to characterize the fungal communities associated with rust pustules from various regions of Costa Rica, to identify regional patterns related to location and altitude, as well as detecting common fungal taxa that may act as mycoparasites of *Hemileia vastatrix*. Beyond their ecological significance, these taxa might hold notable biotechnological potential for sustainable coffee leaf rust management through utilization of produced bioactive compounds with antifungal properties. To our knowledge, this is the first investigation employing ITS metabarcoding to explore the composition and diversity of fungal communities inhabiting coffee leaf rust (CLR) pustules within Costa Rican coffee agroecosystems. This study has the potential to further entangle the structure and responses of fungal communities during CLR infections and further deepen the knowledge of endophytic fungal communities in coffee plantations in Costa Rica and the development of microbiome-informed strategies for integrated disease control in coffee systems.

Materials and methods

Collection of material

A total of 27 uredospore samples were collected from different coffee-growing regions across Costa Rica between (García-Nevarez, et al., 2019) and 2021 (Fig. 1) mostly on the varieties Caturra and Catuaí (Table 1). In addition, samples of rust uredospores were collected from coffee varieties available in the coffee germplasm collection at Centro Agronómico Tropical de Investigación y Enseñanza (CATIE; marked as “Turrialba”). Infected leaves were collected and stored in separate paper bags protected from light.

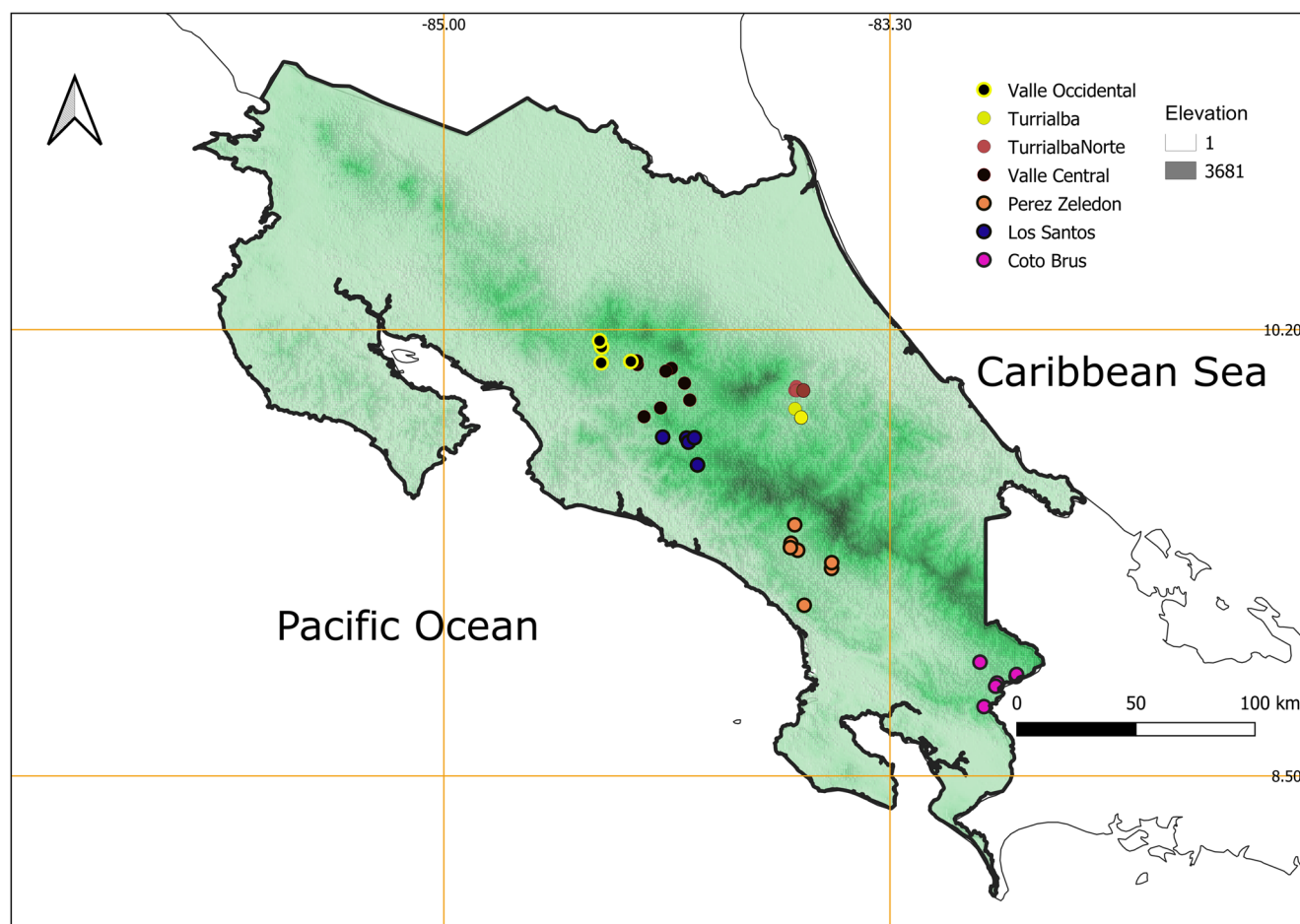


Fig. 1 Geographic distribution of coffee leaf rust (CLR) pustule sampling sites across Costa Rica. Colored points represent the seven sampled coffee-growing regions: Valle Occidental (yellow), Turrialba (light green), Turrialba Norte (brown), Valle Central (black), Pérez

Zeledón (blue), Los Santos (orange), and Coto Brus (magenta). The background grayscale represents the country's elevation gradient, ranging from 1 to 3681 m above sea level

Uredospores were separated in the laboratory by mechanical removal from the leaf using a sterile scalpel knife. The isolated uredospores were deposited in clear gelatin capsules to preserve their viability (Capsule Connection, LLC, Prescott AZ, United States of America), one capsule with uredospores per collection point, consisting of uredospores collected from different individuals. Samples were stored in 32% H_2SO_4 at -20°C .

DNA extraction

Following the protocol of Kosaraju et al. (2017), 20 mg of uredospore material was added into 1.5 ml microcentrifuge tubes, and the extraction buffer (2% CTAB [w/v], 100 mM Tris-HCl [pH=8.0], 25 mM EDTA, 2 M NaCl) was added. Each sample was macerated using a sterile pestle mounted on a handheld drill for 1 min, and 0.1% beta-mercaptoethanol was added to each tube and immediately incubated

at 60°C for 1 h, with moderate shaking (100 rpm). After this, the tubes were allowed to reach room temperature, and centrifuged at 6000 rpm for 20 min, the supernatant was transferred to a new tube where 700 μl of chloroform-isoamyl alcohol (24:1) was added, and the process was repeated twice. The supernatant obtained was transferred to 2 ml tubes and precipitated with 0.7 volumes of isopropanol previously cooled to -20°C , incubated at room temperature for 30 min, and subsequently refrigerated overnight before centrifugation at 14,000 rpm for 20 min at room temperature. The pellet obtained in this step was washed with 75% alcohol for 10 min, centrifuged for 5 min, decanted the excess ethanol, air dried and dissolved in 50 μl of Tris-HCl buffer (10 mM). The DNA obtained was analyzed visually using 1% m/v agarose gels, 100 v, for 55 min, applying GelRed to each sample and placed in a transilluminator. Extracted DNA was stored at -20°C until further use.

Table 1 Metadata of coffee leaf rust samples including sample ID, region, collection date, geographic coordinates, plant variety, and elevation (msl: meters above sea level)

Sample-ID	Region	Sampling date	Latitude	Longitude	Variety	Altitude [msl]
S3_1	Coto Brus	07.02.2019	8.878834	-82.8193650104529	Caturra	977
S1_1	Coto Brus	07.02. 2019	8.844979	-82.8914124933902	Caturra	638
S2_1	Coto Brus	07.02. 2019	8.854023	-82.8913854249514	Catuai- Caturra	638
S4_1	Coto Brus	07.02. 2019	8.933614	-82.9565923735759	Venecia	588
S9_2	Los Santos	22.10. 2019	9.786859	-84.0743437784467	Caturra	1450
S10_3	Los Santos	02.06. 2020	9.787951	-84.044966	Catuai	906
S3_3	Los Santos	19.11. 2019	9.790386	-84.165705	Catuai	977
S7_1	Perez Zeledon	07.09. 2019	9.150405	-83.6263002582707	Venecia	779
S5_1	Perez Zeledon	07.03. 2019	9.290702	-83.5224447899452	Catuai	822
S8_1	Perez Zeledon	07.09. 2019	9.359989	-83.6513326917387	Catuai	586
S2_2	Perez Zeledon	16.10. 2019	9.386823	-83.6771172912937	Catuai	1201
S3_2	Perez Zeledon	16.10. 2019	9.457313	-83.6625857359135	Catuai	1383
S3_5	Turrialba	29.10. 2021	9.898119	-83.661587	CIFC H17	935
S2_6	Turrialba	29.10. 2021	9.898119	-83.661587	5020_CATIE	586
S1_5	Turrialba	29.10. 2021	9.863619	-83.63834	Catuai	1207
S1_6	Turrialba	29.10. 2021	9.863619	-83.63834	Catuai	942
S10_1	Turrialba Norte	17.07. 2019	9.968498	-83.6319570105195	Caturra	1492
S11_1	Turrialba Norte	17.07. 2019	9.982226	-83.6564572499559	Caturra	651
S2_3	Turrialba Norte	11.06. 2019	9.967584	-83.655427	Caturra	1037
S3_6	Valle Central	29.10. 2021	9.931622	-84.062659	Caturra	1000
S9_3	Valle Central	28.01. 2020	9.995887	-84.082459	Catuai	1357
S4_3	Valle Central	17.12. 2019	9.90161	-84.173821	Catuai	1003
S7_2	Valle Occidental	10.02. 2019	10.081218	-84.2646744846061	Catuai	1040
S6_2	Valle Occidental	10.09. 2019	10.078802	-84.2871512533566	Catuai	1510
S5_2	Valle Occidental	10.09. 2019	10.073574	-84.4002770541436	Various	678
S6_3	Valle Occidental	14.01. 2020	10.133793	-84.398369	Caturra	1391
S7_3	Valle Occidental	14.01. 2020	10.157733	-84.406127	Catuai	1323

Sequencing and data analysis

Extracted DNA was sent to NOVOGENE for amplification of the ITS1 region (forward: 5'-GGAAGTAAAAGT CGTAACAAGG-3'; reverse: 5'-GCTGCGTTCTTCATCG ATGC-3'). DNA libraries were sequenced on an Illumina MiSeq in 2×300 bp sequencing mode. Primers and adapters were removed with Cutadapt (Martin 2011) (version 4.2). Trimmed sequences were then merged, quality filtered, and denoised with DADA2 (Callahan et al. 2016) within the Qiime2 toolset (Bolyen et al. 2019) (version 2022.11.1). The chosen cut-off values were 217 bp for forward and 219 bp for reverse reads, respectively. Taxonomic mapping was done with the “vsearch” classifier (Rognes et al. 2016) within Qiime2 against the UNITE 9 database (Kõljalg et al. 2005). Amplicon Sequence Variant (ASV) tables were constructed with the “export” functions of Qiime2 for the taxonomic mapping result and the denoised ASV abundance table (Supplementary Table 1). Functional assignments were done with FungalTraits (Pölme et al. 2020) (version 0.03). Matches to genera were extracted with “merge” in R (Supplementary Table 2).

Rarefaction curves were calculated with the vegan package by utilizing the “rare” and “rarecurve” commands on the obtained ASV table (Oksanen et al. 2024) (version 2.6–6.1) in R (R Core Team 2023; version 4.4.1). Qiime2 output and tables were visualized and further analyzed within R and RStudio (Posit team 2024) (version 2023.12.1), mainly with the microeco package (Liu et al. 2021). Taxonomic differences among groups were tested with the ALDEx2 (Fernandes et al. 2013) module in “kw” configuration, performing a Kruskal-Wallis test. Community testing was done in R with the vegan package by calculating a Bray-Curtis distance matrix (“veg”) and calculating a PERMANOVA on that matrix with “adonis2” (Oksanen et al. 2024). Alpha diversity was calculated after exclusion of mitochondria, chloroplasts, and *H. vastatrix* with the microeco package (utilizing vegan) and tested with the pairwise Wilcoxon test. Ordination analysis was done by using the Bray-Curtis distance matrix for nonmetric multidimensional scaling (NMDS) within ampvis2. For this, the data was rarefied to 13,356 counts per sample. Abiotic measurements were correlated to the ordination by using the “envfit” function within the visualization package ampvis2 (Andersen

et al. 2018) (version 2.8.7). FungalTraits predictions have been visualized with ampvis2 as relative abundances and tested for differences among the regional groups with PERMANOVA based on a Bray-Curtis dissimilarity matrix, calculated with vegan (Oksanen et al. 2024) in R. Elevation data was calculated based on coordinates with elevatr package in R (Hollister et al. 2023).

Results

A total of 3.8 million high-quality sequences were generated across all samples, with an average of 144,298 sequences per sample after filtering steps, representing ~73% of the raw input (Table 2). The rarefaction curves (Supplementary Fig. 1) indicate sufficient sequencing depth for all samples as the curves plateaued, indicating that most fungal diversity was captured. These results supported the reliability of subsequent diversity and community analyses, with differences in curve heights reflecting true biological variation among samples.

As expected, the highest abundance of all taxa was detected for *H. vastatrix* with varying values between 28.0% and 80.4% (Supplementary Fig. 2). Hypocreales was the dominant fungal order across all samples, with relative abundances ranging from 4.4% to 61.8%, although its distribution varied greatly among samples (Supplementary Fig. 3). Interestingly, no Hypocreales were detected in samples from the Valle Central region (Supplementary Fig. 3). At genus level, the most abundant genera were unclassified Pleosporales or Nectriaceae (Fig. 2). In the Coto Brus region, which is the most geographically remote region compared to the central valley, *Setophoma* was more abundant (20.2%), which also includes plant pathogenic species. Significant differences in relative abundance among regions were observed only for unclassified fungal taxa and an unclassified genus within the order Pleosporales ($p < 0.02$; Supplementary Fig. 4). No significant differences in overall community composition were observed among the analyzed regions ($p > 0.05$).

Diversity values differed between regions, albeit not significantly (Fig. 3). Higher Shannon diversity values were

Table 2 Sequencing summary statistics for all CLR pustule samples, including raw reads, quality-filtered, denoised, merged, and non-chimeric read counts with corresponding percentages

Sample-ID	Raw input	Quality filtered	Passed quality filtered [%]	Denoised	Paired-end merging	Merged sequences [%]	Non-chimeric	Non-chimeric [%]
S3_1	204,395	171,188	83.75	170,236	167,574	81.99	154,643	75.66
S1_1	203,309	161,764	79.57	160,631	157,717	77.58	147,145	72.38
S2_1	202,104	176,085	87.13	175,511	173,259	85.73	163,142	80.72
S4_1	204,763	153,176	74.81	151,932	147,780	72.17	139,002	67.88
S9_2	206,288	160,896	78.00	159,897	157,267	76.24	150,853	73.13
S10_3	179,108	150,178	83.85	149,761	148,282	82.79	144,185	80.50
S3_3	213,469	169,442	79.38	168,296	165,052	77.32	157,392	73.73
S7_1	204,473	134,063	65.57	132,632	129,424	63.30	124,626	60.95
S5_1	205,681	161,574	78.56	160,426	156,494	76.09	152,264	74.03
S8_1	200,829	131,696	65.58	130,378	126,691	63.08	120,398	59.95
S2_2	212,957	150,175	70.52	147,932	143,639	67.45	134,351	63.09
S3_2	182,390	124,489	68.25	122,753	118,452	64.94	108,519	59.50
S3_5	147,541	93,256	63.21	93,079	92,760	62.87	91,715	62.16
S2_6	206,364	173,934	84.29	172,885	169,827	82.29	160,192	77.63
S1_5	184,623	158,144	85.66	156,886	153,466	83.12	143,099	77.51
S1_6	202,678	167,111	82.45	166,335	163,988	80.91	155,589	76.77
S10_1	202,774	166,729	82.22	165,474	162,566	80.17	152,127	75.02
S11_1	203,040	155,207	76.44	154,251	151,294	74.51	144,298	71.07
S2_3	203,777	148,960	73.10	147,505	144,822	71.07	138,033	67.74
S3_6	208,053	169,755	81.59	168,852	165,885	79.73	158,484	76.17
S9_3	203,580	114,990	56.48	114,782	113,061	55.54	107,041	52.58
S4_3	205,385	146,524	71.34	145,982	143,082	69.67	134,655	65.56
S7_2	205,571	178,785	86.97	177,959	174,116	84.70	157,701	76.71
S6_2	202,738	166,348	82.05	164,531	160,490	79.16	150,401	74.18
S5_2	209,482	187,835	89.67	186,467	182,874	87.30	175,619	83.83
S6_3	231,769	146,416	63.17	145,558	143,897	62.09	137,816	59.46
S7_3	205,707	149,839	72.84	149,258	147,090	71.50	141,686	68.88

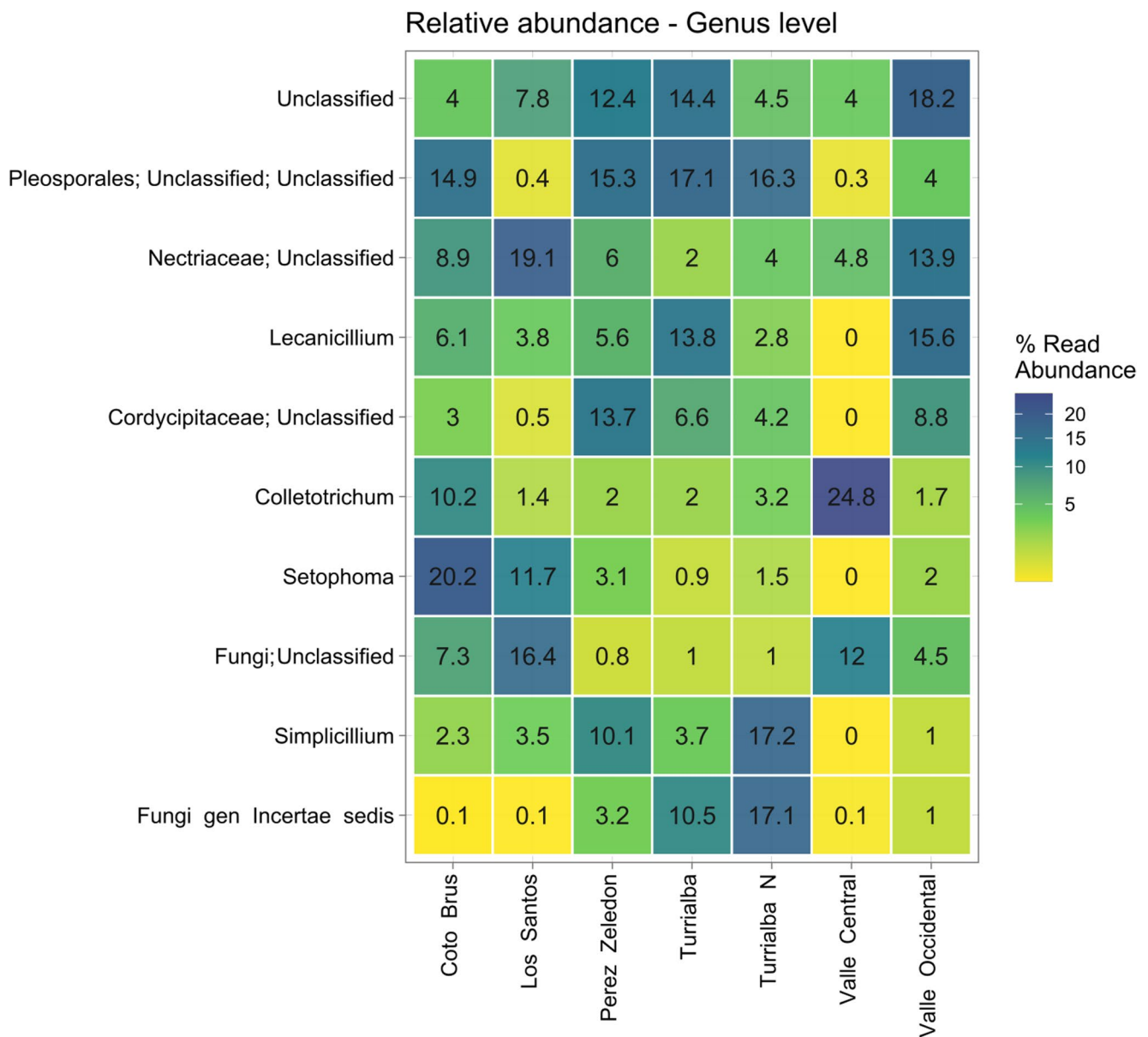


Fig. 2 Heatmap showing the relative abundance (%) of fungal genera associated with coffee leaf rust pustules across seven coffee-growing regions in Costa Rica, excluding *Hemileia vastatrix*. Taxa are ordered

based on highest average abundance across all samples. Color intensity corresponds to read abundance, with darker shades indicating higher relative representation

observed in Pérez Zeledón, Los Santos, and Coto Brus, while the lowest values were found in the Turrialba areas and Valle Central. Valle Occidental exhibited high variance in alpha diversity, resulting in a high mean in comparison, but very strong outliers. Possibly due to the high variability, as seen for the taxonomic profiles, statistical differences among the tested regions were not observed.

NMDS analysis based on a Bray Curtis matrix showed a gradient with distinct clustering of more urbanized regions (Valle Central, Los Santos, Valle Occidental) and more rural and larger plantation areas Turrialba, Turrialba Norte and Pérez Zeledón (Fig. 4). Altitude was identified as a

significant factor influencing the overall fungal community composition ($p < 0.001$), where samples from higher-elevation regions such as Los Santos and Valle Central clustered distinctly from lower-altitude regions like Turrialba and Coto Brus. However, pairwise comparisons did not yield significant results ($p > 0.05$).

Functional predictions with FungalTraits

Functional predictions were calculated with FungalTraits and could classify 12.54% of all taxonomic units on average with the “Unclassified” ranging from 74.59% to 93.34%. *H.*

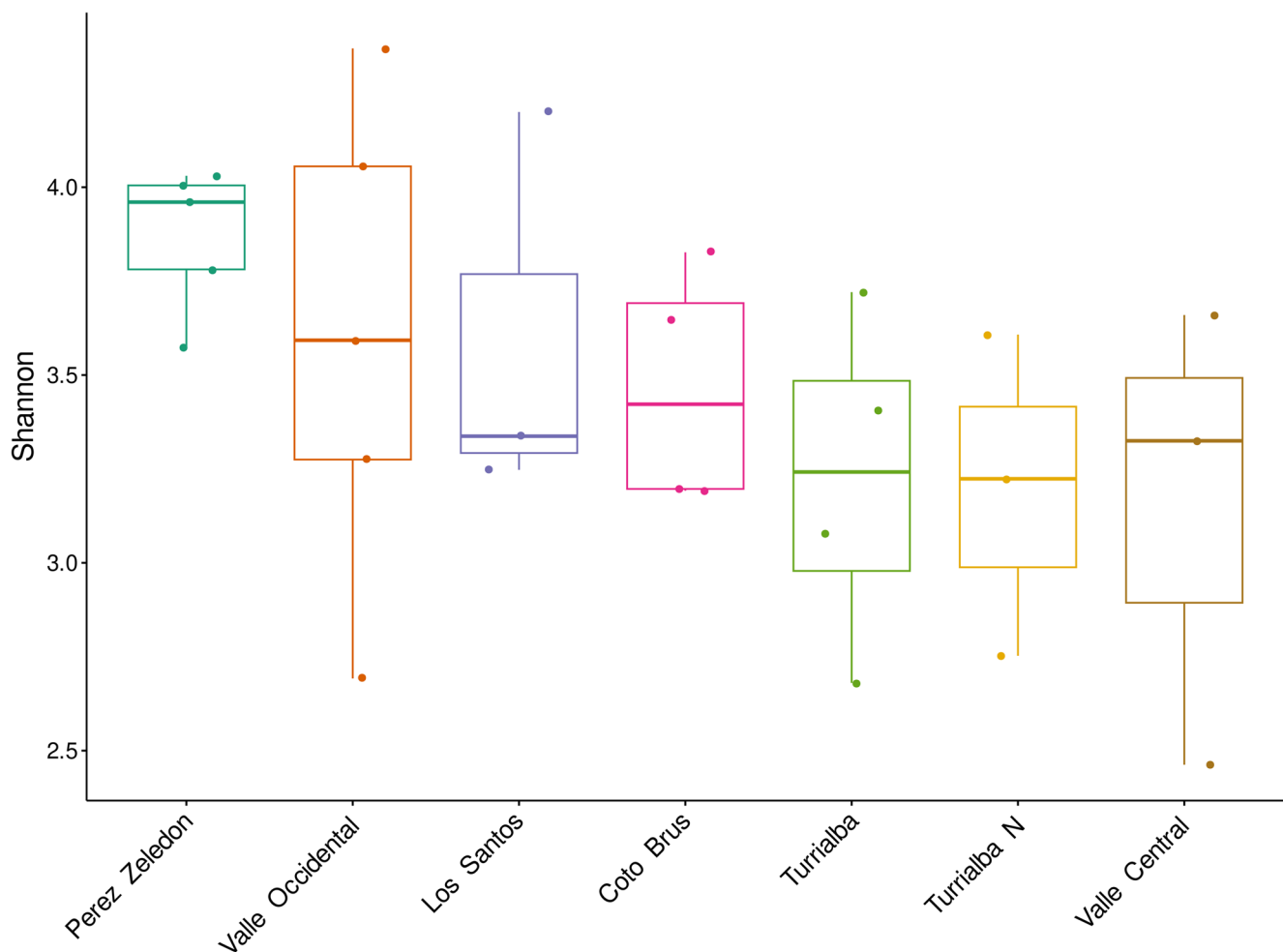


Fig. 3 Alpha diversity of fungal communities associated with coffee leaf rust pustules across seven regions in Costa Rica, measured by the Shannon diversity index. Boxplots represent the median, interquartile range, and individual sample variation within each region

vastatrix counts were filtered before analysis (Supplementary Table 2). Pathotrophs were the most abundant functional group across all samples, with a notable lower abundance in Turrialba (Fig. 5). Notably, Turrialba also showed a higher abundance of Symbiotrophs. Consistent with the environmental fit analysis, elevation was again detected as a significant factor for functional predictions. No significant difference could be measured for the different regions.

Discussion

One of the most relevant findings of this study was the consistent detection of potential mycoparasitic fungi, particularly *Lecanicillium*, across all CLR pustule samples. This genus, known for its entomopathogenic properties, has also been documented as a mycoparasite of *H. vastatrix* (Jackson et al. 2012; James et al. 2016; Liu et al. 2023b; Das et al. 2024; Rissi et al. 2024), and its widespread occurrence suggests a previously underexplored biological interaction

within CLR infections. In addition, members of *Colletotrichum*, which are known to adopt diverse lifestyles ranging from plant pathogens to endophytes (Liu et al. 2023a; Asad et al. 2023), and taxa from the family *Nectriaceae*, were detected frequently. These fungal groups may represent co-infecting organisms, either exploiting weakened host defenses or actively interacting with the rust fungus. Notably, the most abundant trophic group in the functional prediction analysis across all samples was pathotrophs. This supports the idea of co-occurring pathogen infecting the plant or at least being present in some form during rust infection. It is important to note that the high proportion of unclassified taxa and the limited functional annotation coverage reduce the reliability of the functional predictions presented here. Therefore, these results should be viewed as indicative rather than definitive (Zewdie et al. 2020; Daba et al. 2024). Together, these observations highlight a rich and possibly dynamic ecological network within infected tissues, positioning co-occurring fungi as promising targets for future biocontrol research.

NMDS based on Bray Curtis dissimilarity matrix

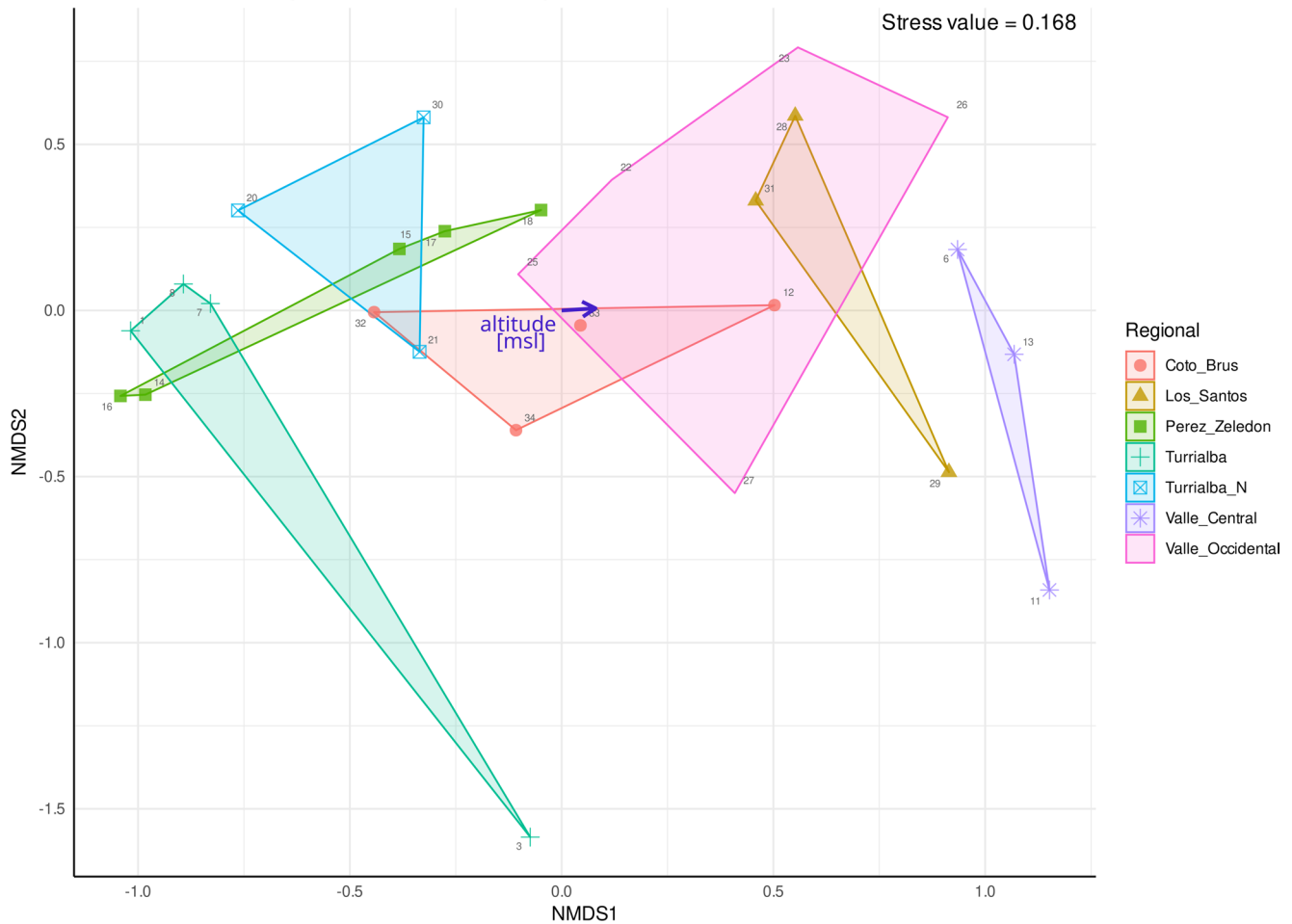


Fig. 4 Non-metric multidimensional scaling (NMDS) plot based on Bray–Curtis dissimilarity of fungal community composition associated with coffee leaf rust pustules across seven regions in Costa Rica.

Each point represents a sample, colored and shaped by region, with polygons enclosing samples from the same region

Although our study did not directly quantify the incidence or severity of CLR across regions, the regional differences observed in fungal community composition may influence local disease dynamics. The recurrent detection of *Lecanicillium* and *Simplicillium*—both previously reported as natural mycoparasites of *H. vastatrix*—suggests that these genera may play an active role in suppressing CLR development. Their presence in regions with comparatively lower rust incidence, such as Turrialba, supports this hypothesis. Future studies integrating longitudinal sampling and disease incidence data are necessary to confirm these relationships and to evaluate which mycoparasitic species are most effective under varying environmental and management conditions.

Beyond these taxonomic and functional signals, the study also revealed expected, yet important, differences in fungal community composition across geographic regions. Despite a high proportion of unclassified taxa - common

in fungal metabarcoding studies (Delgado-Baquerizo et al. 2018; Delgado-Baquerizo 2019; Tedersoo et al. 2020), regional patterns were apparent. For instance, unclassified *Pleosporales* and other unclassified fungal groups showed significantly different abundances between regions such as Valle Central, Los Santos, and Turrialba. These differences may reflect underlying environmental variables, such as land-use history or management intensity (Jurburg et al. 2020; Rusterholz and Baur 2023).

Recent work by García-Nevárez et al. (2025) further supports this hypothesis, as they identified five strains of *Simplicillium lanosoniveum* directly isolated from CLR pustules in Turrialba. Their study provided detailed morphological, physiological, and molecular characterization of these isolates and highlighted their potential role as natural mycoparasites of *H. vastatrix*. Notably, the strains varied significantly in conidia size, radial growth, and production capacity - traits that could influence their ecological

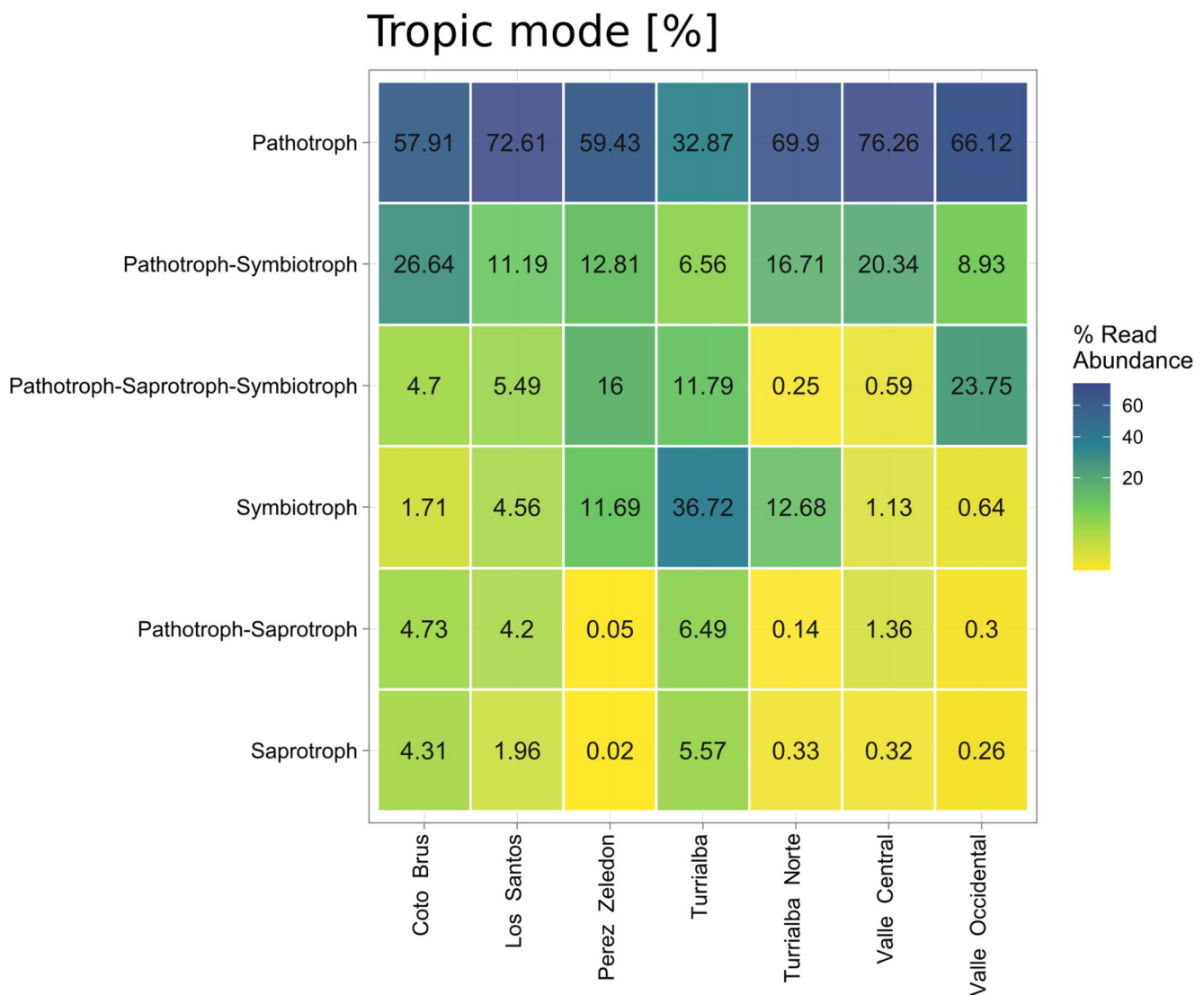


Fig. 5 Relative abundance (%) of fungal trophic modes associated with coffee leaf rust pustules across seven coffee-growing regions in Costa Rica, based on functional predictions using the FungalTraits database

competitiveness and potential efficacy as biocontrol agents. These findings not only confirm the presence of *Simplicillium* in coffee plantations in Costa Rica but also reinforce its relevance in naturally occurring fungal communities during CLR infections. Given that *Simplicillium* was also detected in our community profiles, particularly in regions like Turrialba where CLR incidence is comparatively lower, its activity may reflect active suppression of rust development.

Taxonomic profiles revealed no significant differences between the analyzed regions for the whole communities. When testing genera for differences, only one fully unclassified genus and another unclassified genus belonging to *Pleosporales*, showed significant differences in relative abundance between the regions (with highest differences between Valle Central, Los Santos and Valle Occidental) (Lu et al. 2022; Poma-Angamarca et al. 2024). These regions

are more urban, while the regions with higher relative abundance (Turrialba, Turrialba Norte) are more rural. These results suggest that regional patterns influence co-occurring fungal communities and may also contribute to the dynamics of CLR succession (Rusterholz and Baur 2023).

Diversity showed differences among regions but also with high variances and no clear indication towards a certain type of region (more urban or rural) and not according to the analyzed altitude (with Los Santos and Valle Central having the highest elevation). Given the high dispersion of values, future studies will need to explore the potential role of region and attached physical properties in shaping fungal communities.

Ordination analysis revealed distinct separation between fungal communities in urbanized regions (e.g., Valle Central, Valle Occidental) and those in more rural,

plantation-dominated areas such as Turrialba and Pérez Zeledón. Among the environmental variables tested, altitude emerged as a significant driver of community composition, likely through its effects on microclimatic conditions such as temperature, humidity, and the local prevalence of *H. vastatrix* (Zewdie et al. 2020; Liao et al. 2023; Daba et al. 2024; Tapaça et al. 2024). Even though we detected altitude as a significant factor, the placement and length of the vector indicates that the clustering along the x axis cannot be explained by altitude alone, but other factors that were not identified and tested in this study. Possibly, certain taxa that could not be classified play an important role for this. The stress value of >0.1 also indicates a solid, but not optimal clustering. Notably, regions at higher elevations exhibited a shift in predicted fungal functional profiles, with increased proportions of symbiotrophs and a corresponding decline in pathotrophs—particularly evident in Turrialba. This shift may reflect elevation-driven ecological constraints on the rust pathogen, potentially enabling alternative fungal guilds, including mycoparasites, to proliferate under less favorable conditions for CLR.

In addition to altitude, environmental, and agronomic context may also explain these patterns, as sampling points located near major urban centers contrasted sharply with sites situated within large-scale plantations. Previous studies have shown that agricultural management practices—such as shade cover and historical land use—play a crucial role in shaping microbial communities in coffee systems (Jurburg et al. 2020; Liao et al. 2023). In our study, altitude ranged from 406 to 1,512 m above sea level, introducing variability in factors like temperature and precipitation. These gradients are known to influence both *H. vastatrix* diversity and infection frequency (Zewdie et al. 2020; Liao et al. 2023; Daba et al. 2024; Tapaça et al. 2024). Functional prediction analyses reflected these effects, with the lowest abundance of pathotrophs detected in low-altitude samples from Turrialba. Since rust incidence generally decreases with elevation, it is plausible that mycoparasites or other fungi benefit either from environmental conditions that are more favorable to them or from a decline in CLR competitiveness. These shifts could result from a reactive adjustment of fungal communities to lower pathogen success or from active interactions where mycoparasites suppress CLR under specific altitudinal conditions. Nevertheless, due to the high proportion of unclassified taxa and limited sequence coverage for functional prediction, these interpretations should be approached with caution (Zewdie et al. 2020; Daba et al. 2024).

Conclusion

This study provides the first comprehensive characterization of fungal communities co-occurring with CLR pustules across diverse regions of Costa Rica, revealing that these communities are shaped significantly by both geographic region and altitude. A particularly relevant finding was the widespread presence of potential mycoparasitic fungi, such as *Lecanicillium*, which may play a role in suppressing *H. vastatrix* and represent promising candidates for future biocontrol strategies. While these results underscore the ecological complexity of CLR-infected tissues, a substantial proportion of unclassified taxa highlights a major gap in our taxonomic knowledge of coffee-associated fungi. Future studies should prioritize the identification and functional validation of these unknown taxa and expand regional surveys throughout Latin America to strengthen ecological understanding and identify biotechnological opportunities for CLR management. By establishing a foundational framework for exploring the biotechnological potential of mycoparasitic fungi and associated microbial communities, this work advances the development of sustainable, microbiome-informed approaches for coffee cultivation. Moreover, it paves the way for globally applicable strategies that transcend regional boundaries and contribute to innovative CLR management worldwide.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11274-026-04905-1>.

Acknowledgements A. Gatica-Arias expresses gratitude to the Postdoctoral Program at the University of Costa Rica for its support, as well as to Dr. Dirk Berkelmann for his postdoctoral fellowship.

Authors' contributions A.G.A. and P.C. was responsible for funding acquisition and project administration. A.G.A. and J.M.Z.U conceived and designed the experiments. J.M.Z.U performed the experiments. D.B., J.M.Z.U, P.C., and A.G.A. analyzed and interpreted the data. D.B. wrote the manuscript. All authors reviewed and edited the manuscript.

Funding This work was supported by the University of Costa Rica (Project No. 111-C1-472). This study was also supported in part by the U.S. National Science Foundation grant DEB-1638976 to P. Chaverri.

Data availability Generated sequence data is deposited at the NCBI database under Bioproject ID PRJNA1208051.

Declarations

Competing interests The authors declare no competing interests.

References

- Andersen KS, Kirkegaard RH, Karst SM, Albertsen M (2018) ampvis2: an R package to analyse and visualise 16S rRNA amplicon data. *bioRxiv*. 299537. <https://doi.org/10.1101/299537>
- Anhar A, Abubakar Y, Widayat HP et al (2021) Altitude, shading, and management intensity effect on Arabica coffee yields in Aceh, Indonesia. *Open Agric* 6:254–262. <https://doi.org/doi:10.1515/o-pag-2021-0220>
- Asad S, Priyashantha AKH, Tibpromma S et al (2023) Coffee-Associated Endophytes: Plant Growth Promotion and Crop Protection. *Biology* 12:911. <https://doi.org/10.3390/biology12070911>
- Avelino J, Cristancho M, Georgiou S, Imbach P, Aguilar L, Bornemann G, Läderach P, Anzueto F, Hruska A, Morales C (2015) The coffee rust crises in Colombia and Central America (2008–2013): impacts, plausible causes and proposed solutions. *Food Secur* 7:303–321. <https://doi.org/10.1007/s12571-015-0446-9>
- Bez C, Esposito A, Musonerimana S et al (2023) Comparative study of the rhizosphere microbiome of *Coffea arabica* grown in different countries reveals a small set of prevalent and keystone taxa. *Rhizosphere* 25:100652. <https://doi.org/10.1016/j.rhisph.2022.100652>
- Bilen C, Chami DE, Mereu V et al (2023) A Systematic Review on the Impacts of Climate Change on Coffee Agrosystems. *Plants* 12:102. <https://doi.org/10.3390/plants12010102>
- Bolyen E, Rideout JR, Dillon MR et al (2019) Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol* 37:852–857. <https://doi.org/10.1038/s41587-019-0209-9>
- Breitwieser FP, Lu J, Salzberg SL (2019) A review of methods and databases for metagenomic classification and assembly. *Brief Bioinform* 20:1125–1136. <https://doi.org/10.1093/bib/bbx120>
- Callahan BJ, McMurdie PJ, Rosen MJ et al (2016) DADA2: High-resolution sample inference from Illumina amplicon data. *Nat Methods* 13:581–583. <https://doi.org/10.1038/nmeth.3869>
- Castillo NET, Melchor-Martínez EM, Sierra JSO et al (2020) Impact of climate change and early development of coffee rust – An overview of control strategies to preserve organic cultivars in Mexico. *Sci Total Environ* 738:140225. <https://doi.org/10.1016/j.scitotenv.2020.140225>
- Daba G, Daelemans R, Berecha G et al (2024) Genetic diversity and structure of the coffee leaf rust fungus *Hemileia vastatrix* across different coffee management systems in Ethiopia. *Int Microbiol* 27:525–534. <https://doi.org/10.1007/s10123-023-00409-2>
- Das DK, Machenahalli S, Giri MS et al (2024) Efficacy of biological agent *Lecanicillium lecanii* for the management of coffee leaf rust in India. *Indian Phytopathol* 77:755–762. <https://doi.org/10.1007/s42360-024-00763-9>
- De Oliveira Aparecido L, Torsoni G, Lorençone J, Lorençone P, De Lima R, De Souza Rolim G, Saqui D, De Oliveira G (2024) Future climate suitability of *Hemileia vastatrix* in arabica coffee under CMIP6 scenarios. *J Sci Food Agric*. <https://doi.org/10.1002/jsfa.13838>
- de Resende MLV, Pozza EA, Reichel T, Botelho DMS (2021) Strategies for coffee leaf rust management in organic crop systems. *Agronomy* 11:1865. <https://doi.org/10.3390/agronomy11091865>
- de Sousa LP, Filho OG, Mondego JMC (2021) Differences between the leaf mycobiome of *Coffea arabica* and wild Coffee species and their modulation by caffeine/chlorogenic acid content. *Microorganisms* 9:2296. <https://doi.org/10.3390/microorganism9112296>
- Degnan R, Shuey L, Radford-Smith J, Gardiner D, Carroll B, Mitter N, McTaggart A, Sawyer A (2023) Double-stranded RNA prevents and cures infection by rust fungi. *Commun Biol* 6:1234. <https://doi.org/10.1038/s42003-023-05618-z>
- Delgado-Baquerizo M (2019) Obscure soil microbes and where to find them. *ISME J* 13:2120–2124. <https://doi.org/10.1038/s41396-019-0405-0>
- Delgado-Baquerizo M, Oliverio AM, Brewer TE et al (2018) A global atlas of the dominant bacteria found in soil. *Science* 359 320 LP – 325. <https://doi.org/10.1126/science.aap9516>
- Duong B, Marraccini P, Maeght J-L et al (2020) Coffee microbiota and its potential use in sustainable crop management. A review. *Front Sustain Food Syst* 4:607935. <https://doi.org/10.3389/fsufs.2020.607935>
- Fernandes AD, Macklaim et al (2013) ANOVA-Like Differential Gene Expression Analysis of Single-Organism and Meta-RNA-Seq. *PLoS ONE* 2013 8(7):E67019. <https://doi.org/10.1186/s13059-012-02337-8>
- Fulthorpe R, Martin AR, Isaac ME (2020) Root Endophytes of Coffee (*Coffea arabica*): Variation Across Climatic Gradients and Relationships with Functional Traits. *Phytobiomes J* 4:27–39. <https://doi.org/10.1094/PBIOMES-04-19-0021-R>
- García-Nevarez G, Carnero-Avilés L, Hidalgo-Jamison E (2025) Identification of five strains of *Simplicillium* associated with pustules of coffee rust in Costa Rica. *Revista mexicana de ciencias agrícolas* 16(1):e3392. <https://doi.org/10.29312/remexca.v16i1.3392>
- Gichuru E, Alwora G, Gimase J, Kathurima C (2021) Coffee leaf rust (*Hemileia vastatrix*) in Kenya—A review. *Agronomy* 11:2590. <https://doi.org/10.3390/agronomy11122590>
- Gołębiewski M, Tretyn A (2020) Generating amplicon reads for microbial community assessment with next-generation sequencing. *J Appl Microbiol* 128:330–354. <https://doi.org/10.1111/jam.14380>
- Groenen D (2018) The Effects of Climate Change on the Pests and Diseases of Coffee Crops in Mesoamerica. *J Climatol Weather Forecast* 06:1–5. <https://doi.org/10.4172/2332-2594.1000239>
- Hajian-Forooshani Z, Rivera Salinas IS, Jiménez-Soto E et al (2016) Impact of regionally distinct agroecosystem communities on the potential for autonomous control of the coffee leaf rust. *J Environ Entomol* 45(6):1521–1526. <https://doi.org/10.1093/ee/nvw125>
- Hollister J, Shah T, Nowosad J et al (2023) elevatr: Access Elevation Data from Various APIs. R package version 0.99.1. <https://doi.org/10.5281/zenodo.8335450>. <https://github.com/usepa/elevatr/>
- Jackson D, Skillman J, Vandermeer J (2012) Indirect biological control of the coffee leaf rust, *Hemileia vastatrix*, by the entomogenous fungus *Lecanicillium lecanii* in a complex coffee agroecosystem. *Biol Control* 61:89–97. <https://doi.org/10.1016/j.biocontrol.2012.01.004>
- James TY, Marino JA, Perfecto I, Vandermeer J (2016) Identification of Putative Coffee Rust Mycoparasites via Single-Molecule DNA Sequencing of Infected Pustules. *Appl Environ Microbiol* 82:631–639. <https://doi.org/10.1128/AEM.02639-15>
- Jurburg SD, Shek KL, McGuire K (2020) Soil microbial composition varies in response to coffee agroecosystem management. *FEMS Microbiol Ecol* 96:fiab164. <https://doi.org/10.1093/femsec/fiab164>
- Klindworth A, Pruesse E, Schweer T et al (2013) Evaluation of general 16S ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. *Nucleic Acids Res* 41:1–11. <https://doi.org/10.1093/nar/gks808>
- Köljal U, Larsson K-H, Abarenkov K et al (2005) UNITE: a database providing web-based methods for the molecular identification of ectomycorrhizal fungi. *New Phytol* 166:1063–1068. <https://doi.org/10.1111/j.1469-8137.2005.01376.x>
- Kosaraju B, Sannasi S, Mishra MK et al (2017) Assessment of genetic diversity of coffee leaf rust pathogen *Hemileia vastatrix* using SRAP markers. *J Phytopathol* 165:486–493. <https://doi.org/10.1111/jph.12583>
- Koutouleas A, Collinge D, Boa E (2023) The coffee leaf rust pandemic: An ever-present danger to coffee production. *Plant Pathol* 73(3):522–534. <https://doi.org/10.1111/ppa.13846>

- Laval V, Kerdraon L, Barret M et al (2021) Assessing the Cultivability of Bacteria and Fungi from Arable Crop Residues Using Metabarcoding Data as a Reference. *Diversity* 13:404. <https://doi.org/10.3390/d13090404>
- Liao H, Li C, Ai Y, Li X (2023) Soil bacterial responses to disturbance are enlarged by altitude in a mountain ecosystem. *J Soils Sediments* 23:3820–3831. <https://doi.org/10.1007/s11368-023-03597-6>
- Liu C, Cui Y, Li X, Yao M (2021) microeco: an R package for data mining in microbial community ecology. *FEMS Microbiol Ecol* 97(2):fiaa255. <https://doi.org/10.1093/femsec/fiaa255>
- Liu J-W, Manawasinghe IS, Liao X-N et al (2023a) Endophytic Colletotrichum (Sordariomycetes, Glomerellaceae) species associated with Citrusgrandis cv. Tomentosa China *MycKeys* 95:163–188. <https://doi.org/10.3897/mycokeys.95.87121>
- Liu S, Li J, Feng Q et al (2023b) Insecticidal Effect of the Entomopathogenic Fungus *Lecanicillium araneicola* HK-1 in *Aphis craccivora* (Hemiptera: Aphididae). *Insects* 14:860. <https://doi.org/10.3390/insects14110860>
- Lu L, Karunaratna SC, Dai D-Q et al (2022) Description of Four Novel Species in Pleosporales Associated with Coffee in Yunnan, China. *J Fungi Basel Switz* 8:1113. <https://doi.org/10.3390/jof8101113>
- Luiz BC, Sugiyama LS, Brill E, Keith LM (2024) Survey of potential fungal antagonists of Coffee Leaf Rust (*Hemileia vastatrix*) on *Coffea arabica* in Hawai‘i, USA. *Braz J Microbiol* 55:2839–2844. <https://doi.org/10.1007/s42770-024-01304-2>
- Martin M (2011) Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet J* 17:10–12. <https://doi.org/10.14806/ej.17.1.200>
- Oksanen J, Simpson GL, Blanchet FG et al (2024) vegan: Community Ecology Package. (Version 2.6–8). [R package]. CRAN.
- Oliveira R, Souza R, Lima T, Cavalcanti M (2014) Endophytic fungal diversity in coffee leaves (*Coffea arabica*) cultivated using organic and conventional crop management systems. *Mycosphere* 5:523–530. <https://doi.org/10.5943/mycosphere/5/4/4>
- Phukhamsakda C, Nilsson RH, Bhunjun CS et al (2022) The numbers of fungi: contributions from traditional taxonomic studies and challenges of metabarcoding. *Fungal Divers* 114:327–386. <https://doi.org/10.1007/s13225-022-00502-3>
- Pölme S, Abarenkov K, Henrik Nilsson R et al (2020) FungalTraits: a user-friendly traits database of fungi and fungus-like stramenopiles. *Fungal Divers* 105:1–16. <https://doi.org/10.1007/s13225-020-00466-2>
- Poma-Angamarca RA, Rojas JR, Sánchez-Rodríguez A, Ruiz-González MX (2024) Diversity of Leaf Fungal Endophytes from Two *Coffea arabica* Varieties and Antagonism towards Coffee Leaf Rust. *Plants* 13:814. <https://doi.org/10.3390/plants13060814>
- Posit team (2024) RStudio: Integrated Development Environment for R. Posit Software. PBC, Boston, MA
- R Core Team (2023) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria
- Rissi DV, Ijaz M, Baschien C (2024) Comparative Genomics of Fungi in Nectriaceae Reveals Their Environmental Adaptation and Conservation Strategies. *J Fungi* 10:632. <https://doi.org/10.3390/jof10090632>
- Rognes T, Flouri T, Nichols B et al (2016) VSEARCH: a versatile open source tool for metagenomics. *PeerJ* 4:e2584. <https://doi.org/10.7717/peerj.2584>
- Rusterholz H-P, Baur B (2023) Changes in Soil Fungal Diversity and Composition along a Rural–Urban Gradient. *Forests* 14:2226. <https://doi.org/10.3390/f14112226>
- Schwan RF, Silva C, Batista L (2012) Coffee fermentation. In: *Handbook of Plant-Based Fermented Food Beverage Technology*, 2nd ed.; Taylor & Francis: Boca Raton, FL, USA, 2012; pp 677–690
- Silva Mdo, Guerra-Guimarães C, Diniz L I, et al (2022) An overview of the mechanisms involved in Coffee-Hemileia vastatrix interactions: Plant and pathogen perspectives. *Agronomy* 12:326. <https://doi.org/10.3390/agronomy12020326>
- Strobel G (2018) The emergence of endophytic microbes and their biological promise. *J Fungi* 4:57. <https://doi.org/10.3390/jof4020057>
- Talhinhas P, Batista D, Diniz I et al (2017) The coffee leaf rust pathogen *Hemileia vastatrix*: one and a half centuries around the tropics. *Mol Plant Pathol* 18:1039. <https://doi.org/10.1111/mpp.12512>
- Tapaça Ida, PE, Obieze CC, de Pereira GV M, et al (2024) Irradiance level and elevation shape the soil microbiome communities of *Coffea arabica* L. *Environ Microbiome* 19:75. <https://doi.org/10.1186/s40793-024-00619-9>
- Tedersoo L, Anslan S, Bahram M et al (2020) Identifying the ‘unidentified’ fungi: a global-scale long-read third-generation sequencing approach. *Fungal Divers* 103:273–293. <https://doi.org/10.1007/s13225-020-00456-4>
- Thomas F, Dittami SM, Brunet M et al (2020) Evaluation of a new primer combination to minimize plastid contamination in 16S rDNA metabarcoding analyses of alga-associated bacterial communities. *Environ Microbiol Rep* 12:30–37. <https://doi.org/10.1111/1758-2229.12806>
- Wemheuer F, Berkemann D, Wemheuer B et al (2020) Agroforestry Management Systems Drive the Composition, Diversity, and Function of Fungal and Bacterial Endophyte Communities in Theobroma Cacao Leaves. *Microorganisms* 8:405. <https://doi.org/10.3390/microorganisms8030405>
- Zewdie B, Tack AJM, Adugna G et al (2020) Patterns and drivers of fungal disease communities on Arabica coffee along a management gradient. *Basic Appl Ecol* 47:95–106. <https://doi.org/10.1016/j.baec.2020.05.002>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.